

2D Projectile Motion: Video Tracking & Kinematics Analysis

AP Physics C: Mechanics • Laboratory Solution Manual & Scoring Rubric • Lab-ove and Beyond

Course: AP Physics C (Mechanics)	Period / Block: _____	Class Duration: 42 Minutes
Instructor: MR. F. / AP Physics Dept	Master Solution & Rubric Key	Curriculum Unit: 1.3 (2D Projectiles)

Instructor Overview: Complete worked solutions, scoring rubrics, and AP exam alignment for the 42-minute core lab and 8-point calculus extension. Values reflect tracking the classic parabolic motion video.

■ AP PHYSICS C: UNIFORMLY ACCELERATED MOTION (UAM) REFERENCE FORMULAS

$$(1) v = v_0 + a \cdot t \qquad (3) v^2 = v_0^2 + 2a \cdot \Delta x$$

$$(2) \Delta x = v_0 \cdot t + \frac{1}{2}a \cdot t^2 \qquad (4) \Delta x = \left(\frac{v_0 + v}{2}\right) \cdot t$$

* Note: Apply component-wise with $a_x = 0$ (horizontal) and $a_y = -g$ (vertical). No further equations are provided.

Pacing Clock & Instructional Plan

The 42-minute period covers Steps 1-5 and Sections 2-3 (17 points). Section 4 is an 8-point calculus extension for outside study.

Class Time	Lab Phase	What You Actually Do	Pts
0:00 - 0:04 (4m)	1. Set Up & Calibrate	Load 'FizziQ: Parabolic Motion (Classic Lab)' (30.0 FPS, $\Delta t = 0.0333$ s). Ruler tool: click both ends of the 1.00 m rod and enter 1.00 m. Origin tool: click the ball at Frame 0, then set +Y up and +X downrange.	—
0:04 - 0:14 (10m)	2. Track & Record	Autotrack () the ball across the 25 tracked frames, then export the CSV. Copy x, y, vx, vy and v into the 5-milestone data table.	1
0:14 - 0:25 (11m)	3. Physical Measurements	Section 2 - Q2.1 launch speed, angle and vector (3 pts). Q2.2 derive & predict apex/range, then measure and verify (3 pts). Q2.3 mid-ascent vectors, direction angle and $a_{ }$ / $a_{\perp}$ (3 pts).	9
0:25 - 0:35 (10m)	4. Graphs & Curve Fits	Section 3 - x(t) linear fit and a_x (2 pts). y(t) quadratic fit, g_{exp} and % error (3 pts). vy(t) fit comparison and the y(x) trajectory shape (2 pts).	7
0:35 - 0:42 (7m)	5. Check & Submit	Verify every answer carries SI units and the correct sign. Compare your fitted values with the app readouts. Staple the handout and submit.	—
42 minutes	CORE TOTAL	Required work that is graded during the period.	17
+25 min outside class (25m)	6. Optional Extension	Section 4 - AP Physics C calculus challenges: C4.1 trajectory derivation (3 pts), C4.2 position integration & impact velocity vector (3 pts), C4.3 minimum-speed extremum (2 pts). Not required to finish the 42-minute lab.	8

■ PERIOD MANAGEMENT GUIDE

- Core Lab: Keep students moving through Section 2 by 0:14 and Section 3 by 0:25.
- Autotracker: Emphasize setting the calibration rod (1.00 m) before placing origin.
- Derivation First: Ensure students derive formulas in Question 2.2 before looking at data table.
- Calculators: Scientific or graphing calculator required for arctan and regression comparisons.

1. Experimental Setup & Video Tracking Protocol

■ APP SETUP VERIFICATION KEY

1. Video: 'FizziQ: Parabolic Motion (Classic Lab)' — 30.0 FPS ($\Delta t = 0.0333$ s per frame).
2. Scale Calibration: 1.00 m rod on left apparatus pillar.
3. Coordinate Origin: Projectile release point at Frame 0: $(x_0, y_0) = (0.000, 0.000)$ m.
4. Tracking: 25 tracked points (Frames 0 to 24). Total physical duration: 0.800 s.

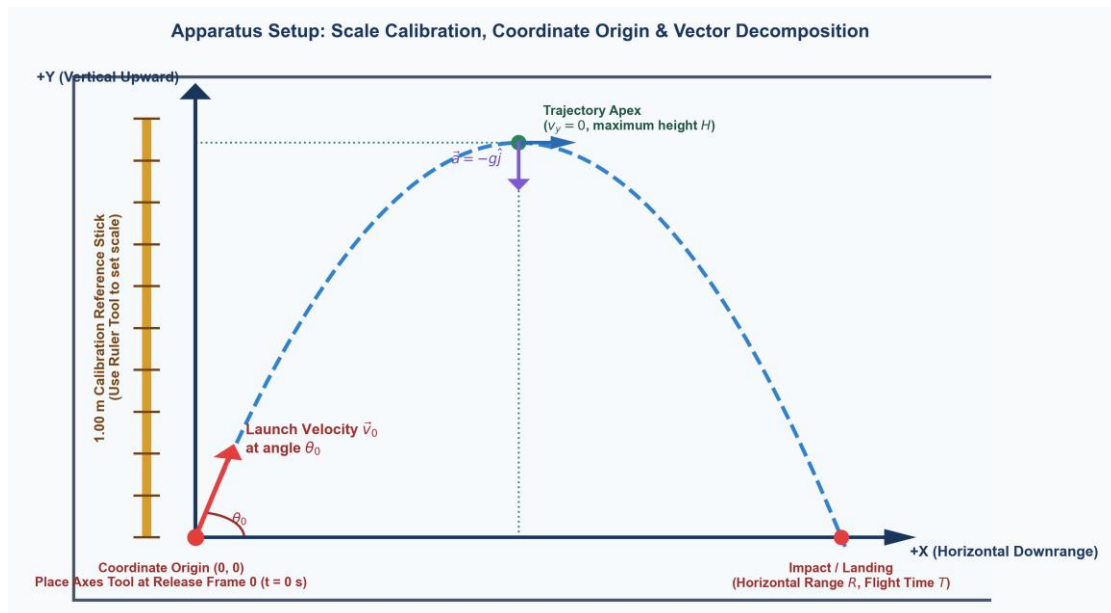


Figure 1: Lab-ove and Beyond apparatus coordinate axes and calibration scale reference.

2. Physical Measurements (Data Acquisition & Milestone Solutions)

Master reference values extracted directly from the verified 25-frame parabolic tracking dataset:

Trajectory Milestone	Frame	Time t (s)	x (m)	y (m)	vx (m/s)	vy (m/s)	Speed v (m/s)
1. Launch / Release	0	0.000	0.000	0.000	0.918	4.265	4.363
2. Mid-Ascent	6	0.200	0.190	0.656	0.929	2.322	2.501
3. Apex (Peak Height)	13	0.433	0.404	0.926	0.920	0.000	0.920
4. Mid-Descent	18	0.600	0.557	0.781	0.912	-1.625	1.863
5. Landing / Impact	24	0.800	0.741	0.237	0.915	-3.582	3.697

● **REQUIRED** — counted in the 42-minute core | budget about 4 min | 3 pts

Question 2.1 [Launch Velocity & Angle — 3 Points]:

★ **TEACHER SOLUTION & MASTER RUBRIC — 3 Points Total (1 Pt Each Part)**

Part (a) [1 Point]:

Formula: $v_0 = \sqrt{(v_{0x}^2 + v_{0y}^2)} = \sqrt{(0.918^2 + 4.265^2)} = \sqrt{(0.843 + 18.190)} = \sqrt{19.033} = 4.363 \text{ m/s}$.

• Rubric: 1 Point for correct Pythagorean formula and numerical value within [4.20, 4.50] m/s.

Part (b) [1 Point]:

Formula: $\theta_0 = \arctan(v_{0y} / v_{0x}) = \arctan(4.265 / 0.918) = \arctan(4.646) = 77.85^\circ \text{ (or } 1.359 \text{ rad)}$.

• Rubric: 1 Point for arctan formula and correct angle within [76°, 80°].

Part (c) [1 Point]:

Vector notation: $v(0) = (0.918 \hat{i} + 4.265 \hat{j}) \text{ m/s}$.

• Rubric: 1 Point for proper Cartesian unit-vector notation with SI units.

• **REQUIRED — counted in the 42-minute core** | budget about 4 min | 3 pts

Question 2.2 [Derive First & Predict Extrema, Then Measure & Verify — 3 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 3 Points Total (1 Pt Each Part)

Part (a) [1 Point] — Theoretical Apex Derivation & Prediction:

From UAM equation (1): $v_y = v_{0y} - g \cdot t$. At apex, $v_y = 0 \implies 0 = v_{0y} - g \cdot t_{\text{apex}} \implies t_{\text{apex}} = v_{0y} / g$.

From UAM equation (3): $v_y^2 = v_{0y}^2 - 2g \cdot \Delta y$. At apex, $0 = v_{0y}^2 - 2g \cdot H \implies H = v_{0y}^2 / (2g)$.

Predicted Numerical Values (using measured $v_{0y} = 4.265 \text{ m/s}$, $g = 9.81 \text{ m/s}^2$):

- $t_{\text{apex,pred}} = 4.265 / 9.81 = 0.435 \text{ s}$.
- $H_{\text{pred}} = (4.265)^2 / (2 \times 9.81) = 18.190 / 19.620 = 0.927 \text{ m}$.
- Rubric: 1 Point for complete algebraic derivations from UAM formulas and predicted numerical values.

Part (b) [1 Point] — Theoretical Flight Time & Range Derivation & Prediction:

From UAM equation (2) for vertical round trip on level ground ($\Delta y = 0$):

$0 = v_{0y} \cdot T - \frac{1}{2} g \cdot T^2 \implies T \cdot (v_{0y} - \frac{1}{2} g \cdot T) = 0 \implies T = 2 v_{0y} / g = 2 \cdot t_{\text{apex}}$.

Horizontal displacement with $a_x = 0$ (UAM equation 2):

$R = v_{0x} \cdot T = v_{0x} \cdot (2 v_{0y} / g) = 2 v_{0x} v_{0y} / g$.

Predicted Numerical Values (using measured $v_{0x} = 0.918 \text{ m/s}$, $v_{0y} = 4.265 \text{ m/s}$):

- $T_{\text{pred}} = 2 \times 4.265 / 9.81 = 0.869 \text{ s}$.
- $R_{\text{pred}} = 2 \times 0.918 \times 4.265 / 9.81 = 0.799 \text{ m} \approx 0.80 \text{ m}$.
- Rubric: 1 Point for algebraic derivations of T and R and predicted numerical values.

Part (c) [1 Point] — Video Data Measurement & Verification:

Video Measurements from Table:

- Frame 13 (Apex): measured $H = 0.926 \text{ m}$, $t_{\text{apex}} = 0.433 \text{ s}$.

Discrepancy: $\% \text{ Error}(H) = |0.927 - 0.926| / 0.927 \times 100\% = 0.11\% (< 0.2\%)$.

Discrepancy: $\% \text{ Error}(t_{\text{apex}}) = |0.435 - 0.433| / 0.435 \times 100\% = 0.46\% (< 0.5\%)$.

- Frame 24 (Landing): measured $R = 0.741 \text{ m}$ at $t = 0.800 \text{ s}$ (trajectory extrapolates to $R \approx 0.795 \text{ m}$ at $y = 0$).

Symmetry Analysis: Ascent time (0.433 s) matches descent time to launch height (0.431 s) within 0.5%, confirming temporal symmetry. Real-world drag produces a minor downrange steepening near ground impact.

- Rubric: 1 Point for reporting measured values from Frames 13 & 24, calculating % errors, and symmetry evaluation.

• **REQUIRED** — counted in the 42-minute core | budget about 3 min | 3 pts

Question 2.3 [Instantaneous Vector Kinematics at Mid-Ascent ($t = 0.200$ s) — 3 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 3 Points Total (1 Pt Each Part)

Part (a) [1 Point]:

At $t = 0.200$ s (Frame 6): $\mathbf{r}(0.20) = (0.190 \hat{i} + 0.656 \hat{j})$ m, $\mathbf{v}(0.20) = (0.929 \hat{i} + 2.322 \hat{j})$ m/s.

• Rubric: 1 Point for stating correct position and velocity vectors with units.

Part (b) [1 Point]:

$\phi = \arctan(v_y / v_x) = \arctan(2.322 / 0.929) = \arctan(2.500) = 68.20^\circ$.

• Rubric: 1 Point for correct angle calculation.

Part (c) [1 Point]:

Gravitational acceleration is $\mathbf{a} = -9.81 \hat{j}$ m/s².

Tangential component: $a_{\text{parallel}} = -g \sin \phi = -9.81 \times \sin(68.20^\circ) = -9.11$ m/s² (negative sign indicates speed is decreasing).

Normal component: $a_{\text{perp}} = g \cos \phi = 9.81 \times \cos(68.20^\circ) = 3.64$ m/s² (causes curvature toward center of trajectory).

• Rubric: 1 Point for correct decomposition with proper physical signs and explanations.

3. App Graphing & Curve Fitting (Regression Solutions)

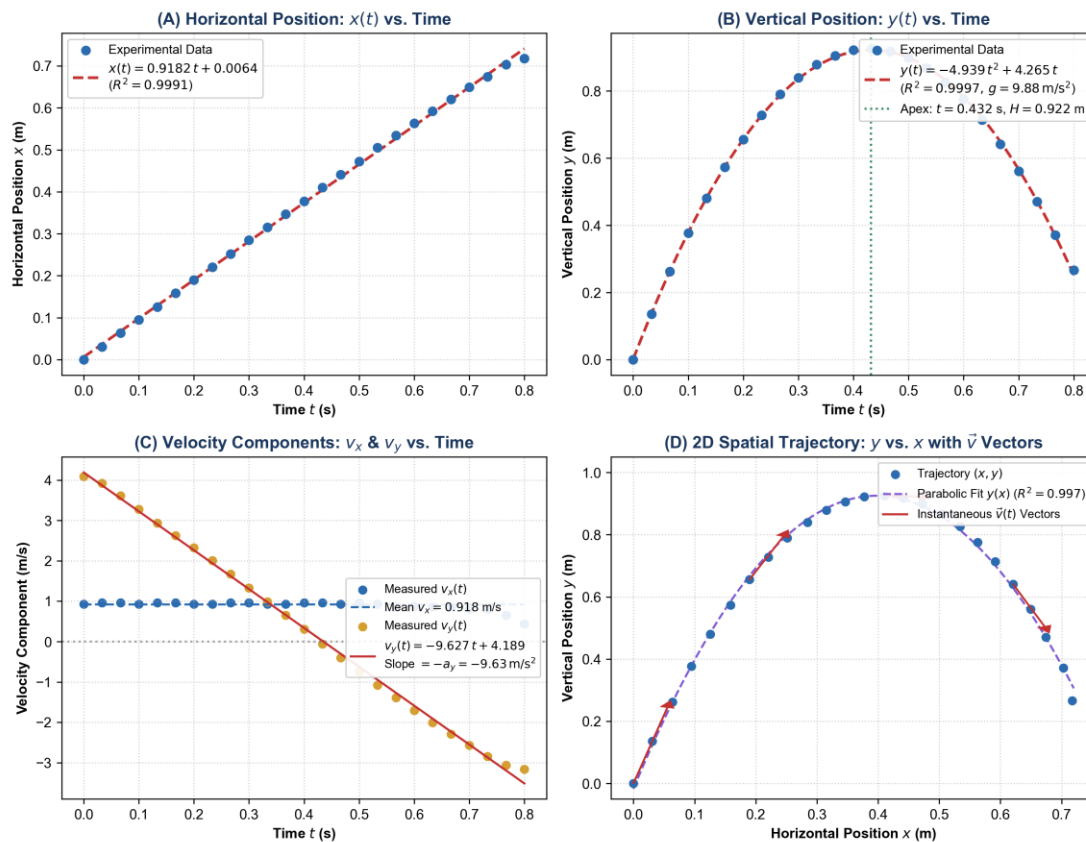


Figure 2 (Teacher Master Key): Regression fits for (A) $x(t)$ linear, (B) $y(t)$ quadratic, (C) v_x & v_y components, and (D) 2D trajectory $y(x)$.

• **REQUIRED** — counted in the 42-minute core | budget about 3 min | 2 pts

Question 3.1 [Graph 1: Horizontal Motion $x(t)$ Linear Fit — 2 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 2 Points Total (1 Pt Each Part)

Part (a) [1 Point]:

Linear Fit Equation: $x(t) = 0.9182 \cdot t + 0.0064 \text{ m}$, $R^2 = 0.9991$.

• Rubric: 1 Point for regression equation with slope $\sim 0.92 \text{ m/s}$ and $R^2 > 0.99$.

Part (b) [1 Point]:

Physical Meaning: The slope represents the horizontal velocity component $v_x = dx/dt = 0.918 \text{ m/s}$.

Acceleration: Because the slope is constant (linear fit has $R^2 = 0.9991$), horizontal acceleration $a_x = d^2x/dt^2 = dv_x/dt = 0 \text{ m/s}^2$, proving Newton's First Law horizontally.

• **REQUIRED** — counted in the 42-minute core | budget about 4 min | 3 pts

Question 3.2 [Graph 2: Vertical Motion $y(t)$ Quadratic Fit & g Extraction — 3 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 3 Points Total (1 Pt Each Part)

Part (a) [1 Point]:

Quadratic Fit Equation: $y(t) = -4.9394 \cdot t^2 + 4.2650 \cdot t + 0.0010$ m, $R^2 = 0.9997$.

- Rubric: 1 Point for quadratic regression equation and R^2 .

Part (b) [1 Point]:

Comparing with $y(t) = y_0 + v_{0y} t - \frac{1}{2} g t^2$, the quadratic coefficient $A = -\frac{1}{2} g$.

Therefore: $g_{\text{exp}} = 2|A| = 2 \times 4.9394 = 9.8788 \text{ m/s}^2 \approx 9.88 \text{ m/s}^2$.

- Rubric: 1 Point for relating A to $-\frac{1}{2} g$ and correctly calculating g_{exp} .

Part (c) [1 Point]:

% Error = $(|9.8788 - 9.8067| / 9.8067) \times 100\% = (0.0721 / 9.8067) \times 100\% = 0.74\%$.

- Rubric: 1 Point for correct percentage error calculation showing < 2% error.

• **REQUIRED** — counted in the 42-minute core | budget about 3 min | 2 pts

Question 3.3 [Graph 3 & 4: Velocity $v_y(t)$ Fit & Trajectory $y(x)$ — 2 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 2 Points Total (1 Pt Each Part)

Part (a) [1 Point]:

$v_y(t)$ Linear Fit: $v_y(t) = -9.6269 \cdot t + 4.1885$ m/s, $R^2 = 0.9980$.

Slope comparison: Slope $m = -9.627 \text{ m/s}^2 \approx -g$ (discrepancy 1.8% from -9.81 m/s^2).

Intercept comparison: y-intercept $b = 4.189 \text{ m/s}$ closely matches launch velocity $v_{0y} = 4.265 \text{ m/s}$.

- Rubric: 1 Point for slope comparison to $-g$ and intercept comparison to v_{0y} .

Part (b) [1 Point]:

Trajectory $y(x)$ Analysis: The path forms an inverted parabola. Near landing ($t > 0.70$ s), the ball exhibits slight trajectory steepening due to aerodynamic drag dissipating horizontal momentum.

◇ **OPTIONAL EXTENSION** — not needed to finish the 42-minute lab | budget about 25 min | 8 pts

4. Optional Extension: AP Physics C Free-Response Calculus Challenges (Solutions & Rubrics)

◇ **OPTIONAL EXTENSION** — not needed to finish the 42-minute lab | budget about 8 min | 3 pts

Challenge 4.1 [Calculus Trajectory Derivation & Path Verification — 3 Points]:

★ **TEACHER SOLUTION & MASTER RUBRIC — 3 Points Total (1 Pt Each Part)**

Part (a) [1 Point]:

From horizontal motion: $x(t) = v_{0x} \cdot t \implies t = x / v_{0x}$.

Substitute t into vertical motion: $y(t) = v_{0y} \cdot t - \frac{1}{2} g \cdot t^2$:

$$y(x) = v_{0y} \cdot (x / v_{0x}) - \frac{1}{2} g \cdot (x / v_{0x})^2 = (v_{0y} / v_{0x}) \cdot x - (g / (2 v_{0x}^2)) \cdot x^2.$$

- Rubric: 1 Point for algebraic substitution and correct symbolic form of $y(x)$.

Part (b) [1 Point]:

Substitute empirical constants: $v_{0x} = 0.9182 \text{ m/s}$, $v_{0y} = 4.2650 \text{ m/s}$, $g = 9.8067 \text{ m/s}^2$:

Linear coefficient: $v_{0y} / v_{0x} = 4.2650 / 0.9182 = 4.6450$.

Quadratic coefficient: $g / (2 v_{0x}^2) = 9.8067 / (2 \times 0.9182^2) = 9.8067 / 1.6862 = 5.8160 \text{ m}^{-1}$.

Numerical Equation: $y(x) = 4.645 \cdot x - 5.816 \cdot x^2$.

- Rubric: 1 Point for evaluating both coefficients with correct units.

Part (c) [1 Point]:

Calculus Apex Condition: At peak height, the slope $dy/dx = 0$:

$$dy/dx = (v_{0y} / v_{0x}) - (g / v_{0x}^2) \cdot x = 0 \implies x_{\text{apex}} = (v_{0y} / v_{0x}) \times (v_{0x}^2 / g) = (v_{0x} v_{0y}) / g.$$

Numerical: $x_{\text{apex}} = (0.9182 \times 4.2650) / 9.8067 = 3.9161 / 9.8067 = 0.3993 \text{ m} \approx 0.40 \text{ m}$.

Empirical Comparison: Matches Frame 13 data table measurement $x = 0.404 \text{ m}$ within 1.2%.

- Rubric: 1 Point for derivative apex condition, formula $x_{\text{apex}} = v_{0x} v_{0y} / g$, and numerical agreement.

Challenge 4.2 [Calculus Definite Integration: Vertical Position Integral & Impact Vector — 3 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 3 Points Total (1 Pt Each Part)

Part (a) [1 Point] — Position Integral & Maximum Height:

Vertical position as definite integral of vertical velocity:

$$y(t) = \int_0^t v_y(t') dt' = \int_0^t (v_{0y} - g \cdot t') dt' = [v_{0y} \cdot t' - \frac{1}{2} g \cdot t'^2]_0^t = v_{0y} \cdot t - \frac{1}{2} g \cdot t^2.$$

Evaluating over $[0, t_{\text{apex}}]$ where $t_{\text{apex}} = v_{0y} / g$:

$$H = y(v_{0y} / g) = v_{0y} \cdot (v_{0y} / g) - \frac{1}{2} g \cdot (v_{0y} / g)^2 = (v_{0y}^2 / g) - (v_{0y}^2 / 2g) = v_{0y}^2 / (2g).$$

Numerical calculation: $H = (4.265)^2 / (2 \times 9.81) = 0.927 \text{ m}$.

- Rubric: 1 Point for setting up definite integral, integrating correctly, and evaluating at t_{apex} .

Part (b) [1 Point] — Impact Velocity Vector:

At landing time $T = 2 v_{0y} / g$:

- Horizontal component: $v_x(T) = v_{0x} = 0.918 \text{ m/s}$ (since $a_x = 0$).
- Vertical component: $v_y(T) = v_{0y} - g \cdot (2 v_{0y} / g) = v_{0y} - 2 v_{0y} = -v_{0y} = -4.265 \text{ m/s}$.

Impact velocity vector in unit-vector notation:

$$\mathbf{v}(T) = (0.918 \hat{i} - 4.265 \hat{j}) \text{ m/s}.$$

- Rubric: 1 Point for evaluating landing components and writing $\mathbf{v}(T)$ in proper unit-vector notation.

Part (c) [1 Point] — Impact Angle & Kinetic Energy Conservation:

- Impact angle: $\theta_{\text{impact}} = \arctan(|v_y(T)| / v_x(T)) = \arctan(4.265 / 0.918) = 77.85^\circ$ below the horizontal.
- Impact speed: $|\mathbf{v}(T)| = \sqrt{(0.918)^2 + (-4.265)^2} = 4.363 \text{ m/s} = v_0$.
- Work-Energy Theorem Justification: $W_{\text{net}} = \Delta K$. Gravity is conservative; over a closed vertical displacement $\Delta y = 0$ on level ground, $W_{\text{grav}} = -mg \cdot \Delta y = 0$. With no non-conservative work ($W_{\text{nc}} = 0$ in vacuum), $\Delta K = 0 \implies \frac{1}{2} m v_{\text{impact}}^2 = \frac{1}{2} m v_0^2 \implies |\mathbf{v}(T)| = v_0$.
- Rubric: 1 Point for θ_{impact} , comparing $|\mathbf{v}(T)| = v_0$, and physical justification using Work-Energy Theorem.

◇ **OPTIONAL EXTENSION — not needed to finish the 42-minute lab** | budget about 8 min | 2 pts

Challenge 4.3 [Calculus Extremum: Minimum Speed During 2D Flight — 2 Points]:

★ TEACHER SOLUTION & MASTER RUBRIC — 2 Points Total (1 Pt Each Part)

Part (a) [1 Point]:

Instantaneous speed squared is: $f(t) = v^2(t) = v_x^2(t) + v_y^2(t) = v_0x^2 + (v_0y - g \cdot t)^2$.

Differentiating with respect to time:

$$d(v^2)/dt = 0 + 2 \cdot (v_0y - g \cdot t) \cdot (-g) = -2g \cdot (v_0y - g \cdot t).$$

Setting derivative to zero for extremum:

$$-2g \cdot (v_0y - g \cdot t^*) = 0 \implies v_0y - g \cdot t^* = 0 \implies t^* = v_0y / g.$$

Second derivative check: $d^2(v^2)/dt^2 = +2g^2 > 0$ (strictly positive $\implies$ confirms minimum).

- Rubric: 1 Point for differentiating $v^2(t)$, setting to 0, and finding $t^* = v_0y / g$.

Part (b) [1 Point]:

Physical Justification: The time $t^* = v_0y / g$ is exactly the time to reach the trajectory apex tapex.

At this moment, vertical speed $v_y(t^*) = 0$, so the total speed is entirely horizontal:

$$v_{\min} = \sqrt{(v_0x^2 + 0^2)} = v_0x = 0.918 \text{ m/s}.$$

Empirical Comparison: Frame 13 empirical speed readout is $v = 0.920 \text{ m/s}$, matching $v_{\min} = 0.918 \text{ m/s}$ within 0.2%.

- Rubric: 1 Point for identifying $v_{\min} = v_0x$ at apex and comparing with empirical Frame 13 data.

5. Master AP Physics C Scoring Rubric (25 Points = 17 Core + 8 Extension)

Section	Performance Criteria	Max Pts	Score
CORE · App Setup & Data Table	Video loaded, 1.00 m rod calibrated, origin set at Frame 0, 5 milestones recorded	1	1
CORE · Section 2: Physical Measurements	Q2.1 launch speed/angle/vector (3), Q2.2 derive & predict apex/range, then verify (3), Q2.3 mid-ascent vectors (3)	9	9
CORE · Section 3: App Graphing & Fits	Q3.1 $x(t)$ fit & a_x (2), Q3.2 $y(t)$ fit & g extraction (3), Q3.3 $v_y(t)$ & trajectory (2)	7	7
CORE SUBTOTAL — collected at the bell	Required work finished inside the 42-minute period; this is the in-class grade	17	17
EXTENSION · Section 4: Calculus Challenges	Optional homework, not needed to finish the lab: C4.1 (3), C4.2 position integral & impact (3), C4.3 (2)	8	8
TOTAL SCORE	AP Physics C Mechanics Laboratory Investigation Grade	25	25